

P2MR Assurance Lab

Independent conformance testing and cross-implementation differential analysis for BIP 360 (Pay-to-Merkle-Root) — concluding in the first independent implementation-conformance report on a quantum-resistant Bitcoin transaction proposal, with Galaxy Research holding first-publication option.

Program Galaxy Bitcoin Quantum Readiness Initiative
Priority area #1 — implementation and review of quantum-resistant transaction proposals
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\$45,000 Phase 1 ask — 12 weeks, 4 payment gates, stop rights at every gate	\$5,000 first decision — retroactive, for a baseline that is already public and already passing	41 checks green 9/9 official vectors + 8/8 boundary tests + 24/24 adversarial mutations, reproducible in ten minutes
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0 Test this application before you read it

This application is designed to be tested, not trusted — because it passes the test.

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# Public baseline — Python 3.10+, zero dependencies, no network, no configuration:
git clone https://github.com/let-the-dreamers-rise/p2mr-assurance-lab
cd p2mr-assurance-lab && python verify_vectors.py && python run_mutations.py

# Output on a fresh clone:
9/9 official BIP 360 construction vectors PASS (byte-for-byte)
8/8 local boundary tests PASS
24/24 adversarial mutation cases PASS
pinned: bitcoin/bips @ ed4ffcb6a48d4dc4fd4fc11cdba783c233db8c66e
fixture: p2mr_construction.json · sha256: cd02da0b3c5bcbea98ed4d3141ebaf71344ee5acc8eb543841b7d2ed853251c7
```

Those five minutes replace every credential this document could otherwise have claimed. The fixture is a byte-identical copy of the official BIP 360 construction vectors at a pinned bitcoin/bips commit; both runners refuse to execute at all if the fixture's SHA-256 deviates from the pin, and continuous integration re-runs the full suite on every push. Everything else in this proposal is the plan for turning that verified baseline into infrastructure the Bitcoin ecosystem needs before BIP 360 reaches activation — and that Galaxy's initiative was created to fund.

1 The gap this lab closes

Every P2MR implementer tests their own code. Nobody is funded to break all of them, impartially, against a pinned specification. That verification gap sits directly on Bitcoin's quantum-migration path.

Galaxy's announcement states the strategic picture precisely: the CRQC timeline is compressing, NIST's post-quantum standards (FIPS 204 and 205) have been final since August 2024, a federal deadline stands at 2031 — and Bitcoin's

decentralized governance means protocol upgrades “can take years to design, review, test and deploy.” BIP 360 (P2MR) is the leading concrete response: a draft output type by Hunter Beast, Ethan Heilman, and Isabel Foxen Duke that removes Taproot’s quantum-vulnerable key path while preserving its script machinery, giving long-exposure protection now and a practical home for post-quantum signature opcodes later.

That design inherits Taproot’s most unforgiving layer: BIP 340 tagged hashing, lexicographically sorted branch pairs, CompactSize serialization, control-block encoding with a mandatory parity bit, leaf-version masking, Bech32m addresses, a hard 128-level depth limit. A single wrong byte anywhere in that pipeline produces an address that accepts funds and can never release them — and produces it silently. Multiple independent implementations already exist: the reference implementation inside the bips repository, an open bitcoinjs-lib branch, and public Bitcoin Core regtest work. Independent implementations are exactly how drafts mature, and exactly where byte-level divergence breeds.

The economics are asymmetric in Galaxy’s favor. A conformance defect caught today, while BIP 360 is a Draft, costs a maintainer a code review. The same defect caught after activation costs users real funds, costs the proposal its credibility, and costs the ecosystem years — the one resource the quantum timeline does not offer. Funding adversarial verification at the draft stage is the cheapest risk reduction available anywhere on the quantum-readiness map, and it is priority area #1 of Galaxy’s own program, verbatim: “implementation and review of quantum-resistant transaction proposals.”

2 What already exists — built before asking

This lab did not wait for funding to start the work. The baseline shipped first, unpaid, because the gap is real — the grant scales it.

The public repository already demonstrates, executably, every competence the program requires:

ALREADY DEMONSTRATED	WHAT IT PROVES
9/9 official vectors, byte-for-byte	Independent, from-scratch implementation of the full construction pipeline — tagged hashes, tree building, scriptPubKey, Bech32m addresses, control blocks — agreeing with the official fixture on every published field, including depth-first leaf-hash ordering.
Both misuse vectors refused with the exact specified errors	The specification's negative space is implemented, not just its happy path: internal pubkeys and empty trees are rejected with the fixture's error strings verbatim.
The misuse vector's secp256k1 bytes independently recomputed	The fixture documents the tweak a faulty P2TR-style construction would produce. The verifier reimplements the elliptic-curve tweak from scratch — solely to prove those exact bytes and confirm P2MR must never produce them.
Spend-side validation walk for every leaf of every vector	Construction is checked in both directions: each (script, control block) pair is walked back through BIP 360's validation rules to the committed witness program — the same computation a consensus implementation performs.
8/8 boundary tests at the edges vectors don't reach	CompactSize encoding boundaries, odd leaf-version rejection, the $m = 128$ depth limit accepted and $m = 129$ refused, branch-sort commutativity, checksum corruption, strict program length, SegWit version discrimination.
24/24 adversarial mutations, each carrying its specification citation	The seed of the funded corpus, already running: merkle-path byte flips, control-block truncation and extension, parity-bit clearing, leaf-version mismatches, script tampering, path reordering, scriptPubKey shape violations, and address-encoding attacks including the legacy-bech32-constant case. Every case names the exact specification sentence that defines its expected outcome.
Version-pinned by construction	BIP 360 is an evolving Draft, so an unpinned test claim is worthless. Both runners hard-refuse to execute against any fixture whose SHA-256 deviates from the recorded pin — and the fixture's bytes are protected against platform line-ending mangling so the check holds on every operating system a reviewer might use.

Repository: github.com/let-the-dreamers-rise/p2mr-assurance-lab · MIT · CI runs the complete suite on every push · this proposal is archived inside it at [docs/galaxy-proposal.pdf](#).

3 What USD 45,000 delivers

1. Pinned adversarial conformance corpus — 250+ cases	The 24-case launch corpus grown roughly tenfold across every construction and validation surface, each case carrying expected outcome, specification citation, and a minimized reproducer. Any present or future P2MR implementer — wallet library, custodian stack, or Core patch — imports it as their conformance gate. The corpus outlives this program and compounds in value with every new implementation.
2. Cross-implementation differential matrix	The full corpus executed against the bips reference implementation, the open bitcoinjs-lib P2MR branch, and available Bitcoin Core regtest patches — each pinned to exact upstream commits, each result strictly pass / fail / not-implemented / not-buildable / not-applicable. Divergence between independent implementations is where stranded-funds bugs live; every security-relevant finding reaches maintainers through coordinated disclosure before it reaches the public.
3. Property-based and fuzz harness, in CI	Continuous randomized construction/parse round-trips with recorded seeds, so conformance is a standing guarantee rather than a one-time snapshot. CI records BIP commit, fixture hash, platform, and counts on every run — no result exists without its provenance.
4. Offline exposure-classifier CLI (v1)	A local-only, zero-telemetry tool classifying output scripts by quantum-exposure class from non-secret exports — the question every custodian will ask (“what fraction of our UTXOs have exposed keys, and what routes preserve our signing policy?”), answered on their own hardware. It hard-refuses seeds, xprv, WIF, and wallet databases, and those refusals are themselves tested. This is the seed of the Phase 2 migration analyzer.
5. “State of BIP 360 Implementation Conformance” — the report	A publication-grade, end-to-end reproducible report: methodology, corpus coverage, the differential matrix, divergence analysis, disclosure outcomes, witness-weight observations, and open questions for the draft's evolution. Galaxy Research receives pre-publication review and first-publication option — the first independent conformance assessment of any quantum-resistant Bitcoin transaction proposal, feeding the initiative's research-and-publishing pillar directly.

4 Why this grant, from your side of the table

A three-week-old program's first funded projects define its reputation. This one is small, fast, already running, impossible to be embarrassed by — and it produces a citable artifact with Galaxy's name on it.

- **You are extending a running project, not underwriting an intention.** The baseline passes on a fresh clone today. MO's \$5,000 is retroactive — the only grant structure in which the first payment follows delivered, independently verifiable work rather than preceding it.
- **The deliverables serve the initiative's own three pillars.** The corpus and matrix are priority-area-#1 developer funding; the conformance report feeds Galaxy Research's publishing program with a first-of-kind artifact; and the findings give the Quantum Advisory Council concrete implementation evidence to weigh, not survey opinions.
- **Independence is structurally credible here.** This lab has no stake in any implementation, no employer, and no reputation to protect except the accuracy of its published results. Differential testing performed outside every implementing team is the strongest form of the evidence — and the form implementers cannot produce for themselves.
- **The downside is capped and administered cheaply.** Four gates, each reviewable in under an hour against a one-page acceptance sheet that maps every criterion to a command, a file, or a diff. Stop at any gate; maximum exposure to a wrong call about the applicant is one tranche.

- **The diligence path costs one email.** Forward this PDF to the BIP 360 authors or any implementation maintainer and ask whether an independent, version-pinned conformance corpus and differential matrix would be useful right now. Their answer is a stronger signal than any paragraph of self-description — and this proposal is confident enough in the answer to invite the question.
- **Cost basis works for you, not against quality.** Twelve weeks of disciplined, full-time, single-owner engineering at an Indian cost basis delivers a multiple of the artifact-per-dollar of any North American consultancy — with every line of output public and pinned, so the quality claim is checkable rather than asserted.

5 The reviewer's checklist — every question, pre-answered

WHAT YOU NEED TO BE TRUE	WHY IT MATTERS	HOW YOU VERIFY IT IN MINUTES
The applicant can implement P2MR's machinery correctly	Core competence for every workstream	Run the \$0 commands: 41 checks, byte-for-byte against the official pinned fixture, on your machine.
The tests probe real failure modes, not padding	Corpus value is per-case, not per-count	Open <code>mutations/corpus.json</code> : every case names its mutation, expected outcome, and the specification sentence requiring it. Sample any five.
Results cannot rot as the Draft evolves	BIP 360 is a moving target	Both runners hard-refuse unpinned fixtures; every artifact names its bips commit and SHA-256; the funded version-diff tracker reports what each draft revision changes.
Money maps to work	Grant hygiene	Payment only follows gate acceptance; a public per-category ledger updates at every gate; the budget has three lines and no vague category.
There is zero safety or regulatory surface	Downside control for Galaxy	No key material, custody, signing, broadcasting, telemetry, or mainnet flows — and the secret-input refusals are tested deliverables, not policy promises.
The applicant will not overclaim near Galaxy's name	Your brand stands near this work	The \$8 banner ships verbatim on every artifact; any release naming Galaxy is submitted to you for review before publication. The repository's README already carries the banner today.

6 The twelve weeks, technically

M1 (weeks 1–3) — corpus to 100+, foundations hardened

Expand the launch corpus past 100 cases across a published mutation taxonomy: construction-side (tree shape, CompactSize edges at every threshold, leaf-version space, duplicate and pathological trees), validation-side (control-byte space, path-length arithmetic, walk-order properties), and encoding-side (Bech32m error classes, program-length space, version discrimination). Ship the threat model, the coordinated-disclosure policy, and a clean-clone reproducibility report covering Linux, macOS, and Windows.

M2 (weeks 4–8) — 250+ cases and the differential matrix

Pin exact upstream commits of the reference implementation, the bitcoinjs-lib branch, and available Core regtest patches; build each reproducibly; execute the full corpus against all targets through thin adapters; and publish the matrix with a minimized reproducer for every divergence. Stand up the property/fuzz harness in CI with recorded

seeds. Any security-relevant divergence goes to maintainers first — findings are published as facts with versions, never as alarm.

M3 (weeks 9–12) — classifier and the report

Deliver the offline exposure-classifier v1 with its refusal tests, then the report: methodology, coverage map, matrix results, divergence analysis and disclosure outcomes, witness-weight observations for P2MR spends, and open questions the draft's evolution must answer — reproducible end-to-end from public artifacts. Galaxy reviews before publication; pins and corpus are maintained for six months after M3 at no further cost.

7 Milestones, payments, stop rights

GATE	WEEK	PAYMENT	CUMULATIVE	ACCEPTED AGAINST
M0	—	\$5,000	\$5,000	The already-public baseline: 41 passing checks, pinned manifest, scope charter, opening funds ledger. Retroactive — you pay only for work you have already verified.
M1	3	\$10,000	\$15,000	100+ case corpus with citations and reproducers; threat model; disclosure policy; three-platform reproducibility report.
M2	8	\$15,000	\$30,000	250+ case corpus; pinned differential matrix across all three targets; fuzz harness in CI; disclosure log for any divergence found.
M3	12	\$15,000	\$45,000	Exposure-classifier v1 with tested refusals; the conformance report, Galaxy-reviewed, reproducible end-to-end; six-month maintenance commitment begins.

Budget: \$36,000 lead engineering (12 weeks, full-time, sole owner) · \$6,000 CI, fuzzing, and multi-platform build infrastructure · \$3,000 disclosure and release operations. Three lines, no vague category. Grant funds never touch trading, tokens, custody, marketing, or undisclosed subcontractors. **Stop rights:** accept, reject, pause, resize, or terminate at any gate; nothing advances automatically; a blocked milestone is reported as blocked, never relabeled complete. **Automatic payment freeze** on any unpinned result, any unreproducible claim, any compatibility statement without an exact tested version, or any undisclosed security-relevant finding.

8 Claim discipline — enforced, not promised

This banner ships verbatim on every artifact this grant funds — it is in the repository today:

“BIP 360 (P2MR) is a Draft and is not activated on Bitcoin mainnet. P2MR can mitigate long-exposure (revealed-key) risk only under fresh-key discipline; it does not by itself defend the reveal-to-confirmation window, and nothing in this work makes Bitcoin quantum-safe today.”

- No consensus change, no activation claim, no representation as a Bitcoin Core release; no token, bridge, or trading product; no seed or private-key handling, custody, signing, broadcasting, telemetry, or mainnet flows — ever.
- “Audit” is reserved for qualified external auditors under contract; this lab's own work is testing, and is named as such.
- Coordinated disclosure, in order: pin and reproduce → minimize → notify maintainers privately → allow a remediation window → publish facts with versions → retest the full corpus. A compatibility bug is never converted into a claim that Bitcoin is broken.

- **Draft-evolution risk, answered:** if BIP 360 is materially revised, the pinned corpus and version-diff tracker retarget to the successor revision — the methodology is proposal-portable. If the proposal is abandoned outright, you stop at the next gate; that is what the gates are for.

9 The applicant, plainly

Credentials are the weakest instrument for evaluating this application, because it offers a stronger one: work you can execute.

Ashwin Goyal — B.Tech computer science undergraduate at VIT Bhopal, India; open-source engineer with roughly ninety public repositories, including a deterministic transaction-policy engine with independently verifiable signed receipts and a live milestone-gated oracle demo. What is not claimed, so you never have to discount for it: no Bitcoin Core commits, no published cryptography research, no prior institutional grants beyond one \$200 developer microgrant. The structure makes those absences immaterial — you never pay ahead of verified work, and the verified work already exists. Full-time, single-owner focus for twelve weeks is the delivery model; the public commit history is the accountability model; the gates are the enforcement model.

10 Phase 2 — earned, not assumed

If Phase 1 delivers, a separately gated Phase 2 proposal will follow in the USD 150,000–300,000 range: the exposure classifier grown into a production-grade wallet/custodian migration-readiness analyzer; descriptor and PSBT preflight tooling (regtest and signet only); a measured FIPS 204 (ML-DSA) / FIPS 205 (SLH-DSA) witness-cost benchmark under current consensus limits; and an externally reviewed Quantum Migration Reference Architecture. Nothing in Phase 2 is part of this ask — and Phase 1 stands complete on its own: the corpus, the matrix, and the report retain their full value whether or not Phase 2 ever happens.

11 References and closing

Galaxy — Bitcoin Quantum Readiness Initiative announcement, 2026-07-21 · bitcoin/bips — BIP 360 (P2MR), pinned at ed4ffcb6a48d4dc4fd4c11cdba783c233db8c66e; BIP 340; BIP 341; BIP 174; BIP 3 · NIST — FIPS 204 (ML-DSA), FIPS 205 (SLH-DSA), August 2024 · Bitcoin Optech — quantum-resistance topic · Delving Bitcoin — BIP 360 threads · public P2MR implementation surfaces: bips reference implementation, open bitcoinjs-lib pull request, independent Bitcoin Core regtest work (pinned per-target at M2).

The baseline is public. The suite is green. The command is one line. Run it, sample the corpus, forward this document to a maintainer you trust — this application welcomes every form of scrutiny you can apply, because scrutiny is the product it is asking you to fund.

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